

Linear/Matrix Transformations

Linear or Matrix Transformation

Transformation is essentially a fancy word for function; it's something that takes in inputs, and spits out some output for each one. Specifically, in the context of linear algebra, we think about transformations that take in some vector and spit out another vector.

Linear Transformation

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Input

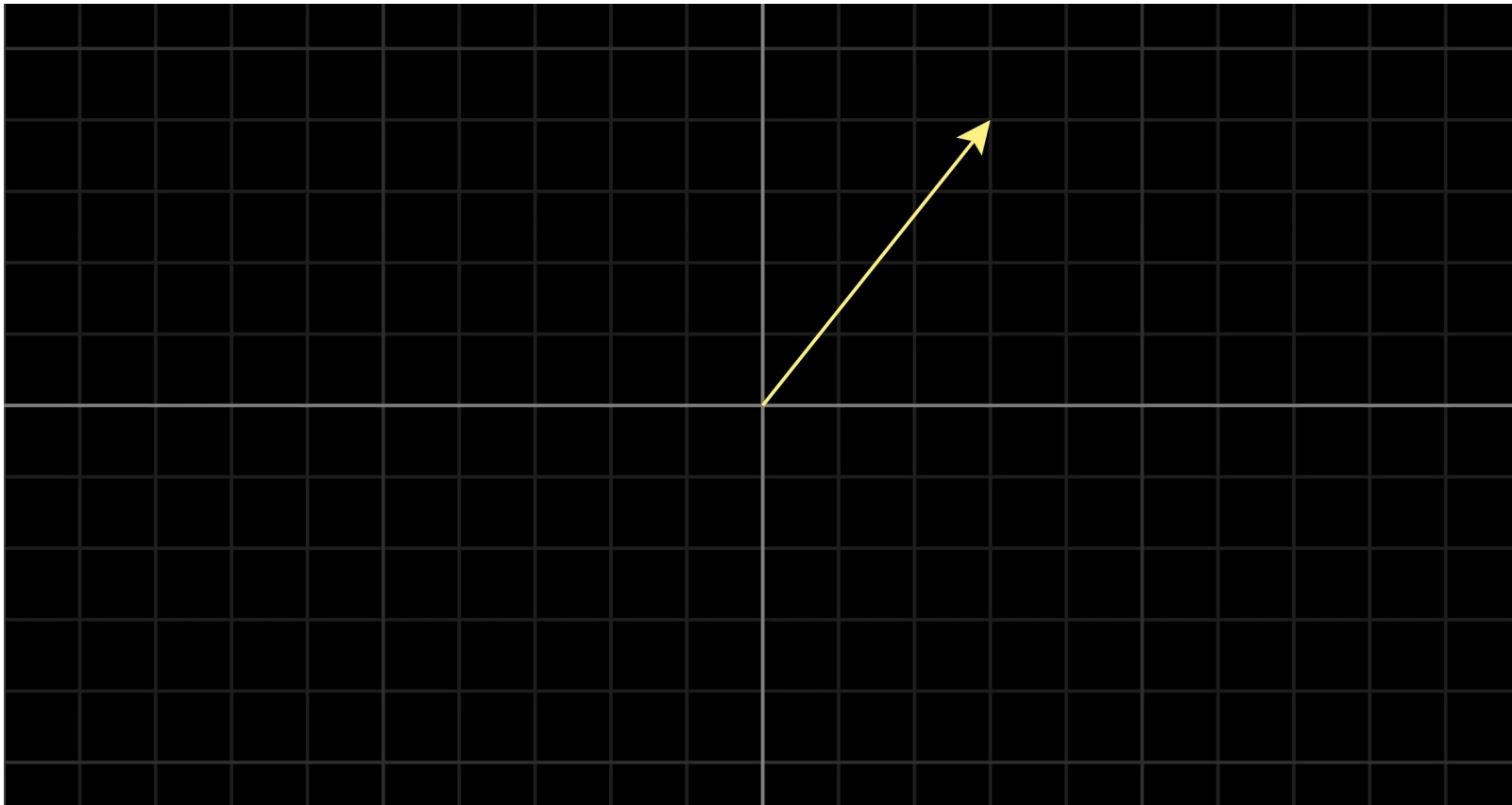
$L(\vec{v})$

$$\begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

Output

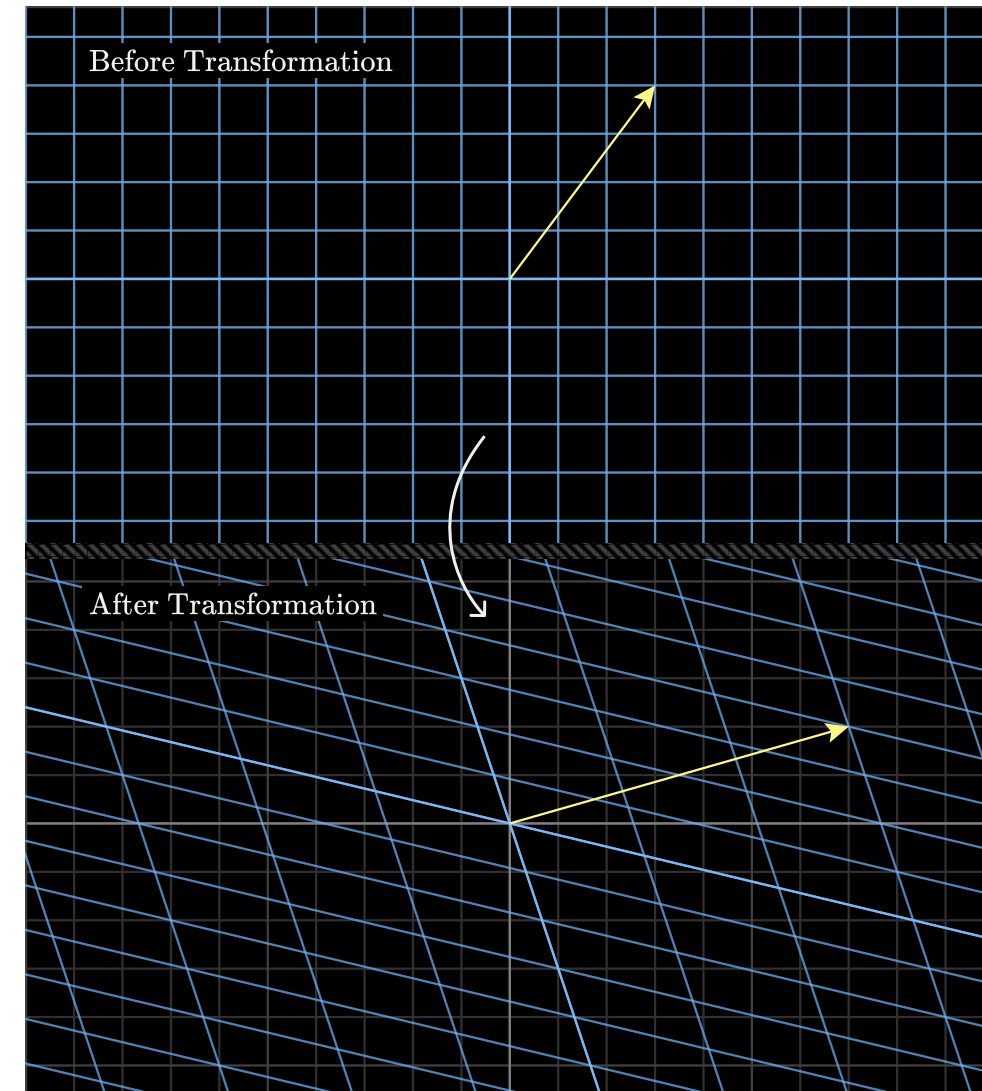
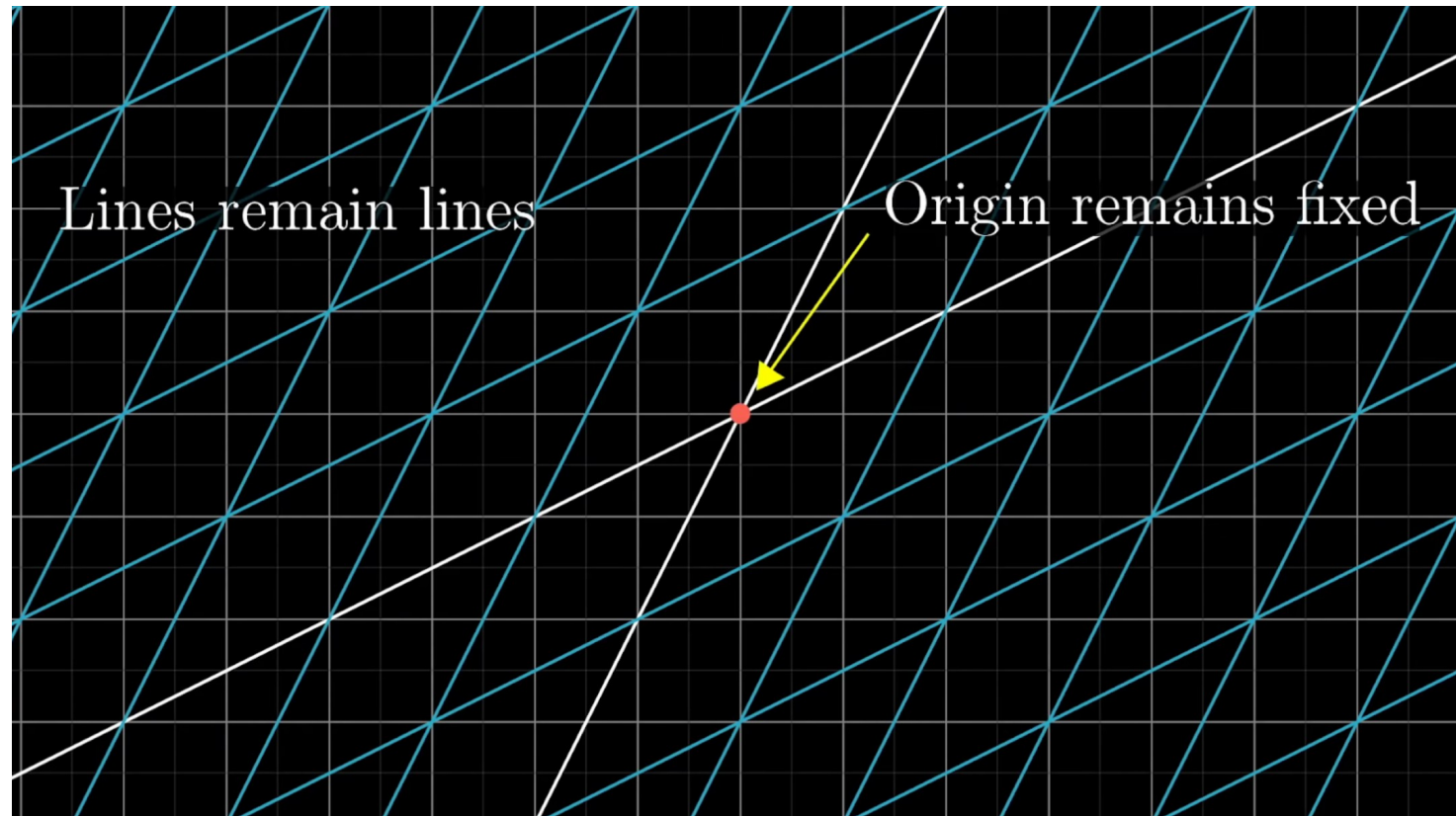
Linear or Matrix Transformation

So why use the word "transformation" instead of "function" if they mean the same thing? It's to be suggestive of a certain way to visualize this input-output relation. Rather than trying to use something like a graph, which really only works in the case of functions that take in one or two numbers and output a number, a great way to understand functions of vectors is to use *movement*.



What makes a transformation "linear"?

A transformation is "linear" if it has two properties: all lines must remain lines, without getting curved, and the origin must remain fixed in place.



Linear Transformation

A *matrix transformation* $T: R^n \rightarrow R^m$ is a function of the form

$$T(\vec{x}) = A * \vec{x}$$

for all $\vec{x}$ vectors in R^n and an $m \times n$ matrix A .

The matrix transformations are precisely the *linear transformations* from R^n to R^m , that is, the transformations with the linearity properties

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(k\mathbf{u}) = kT(\mathbf{u})$$

where $u, v \in R^n$ and $k \in R$

Types of Linear Transformation

There are two types of linear transformations (defining from $\mathbb{R}^2$ to $\mathbb{R}^2$):

1. Euclidean Transformation
2. Affine Transformation

Euclidean Transformation

A **Euclidean transformation** is a geometric transformation that preserves the distances and angles between points. These transformations include:

Translations: Shifting all points in a space by the same distance in a specific direction.

Reflections: Flipping points over a line (in 2D) or a plane (in 3D).

Rotations: Turning points around a fixed center.

Translation of Point

Translation is a transformation that moves every point of a shape or object by the same amount in a given direction.

In 2D, translating a point (x, y) by a vector $\vec{t} = (a, b)$ means shifting:

$$(x, y) \longrightarrow (x + a, y + b)$$

Example: (Translation of a triangle)

Let $A = (-2, -2)$, $B = (2, -2)$, $C = (0, 2)$ form a triangle.

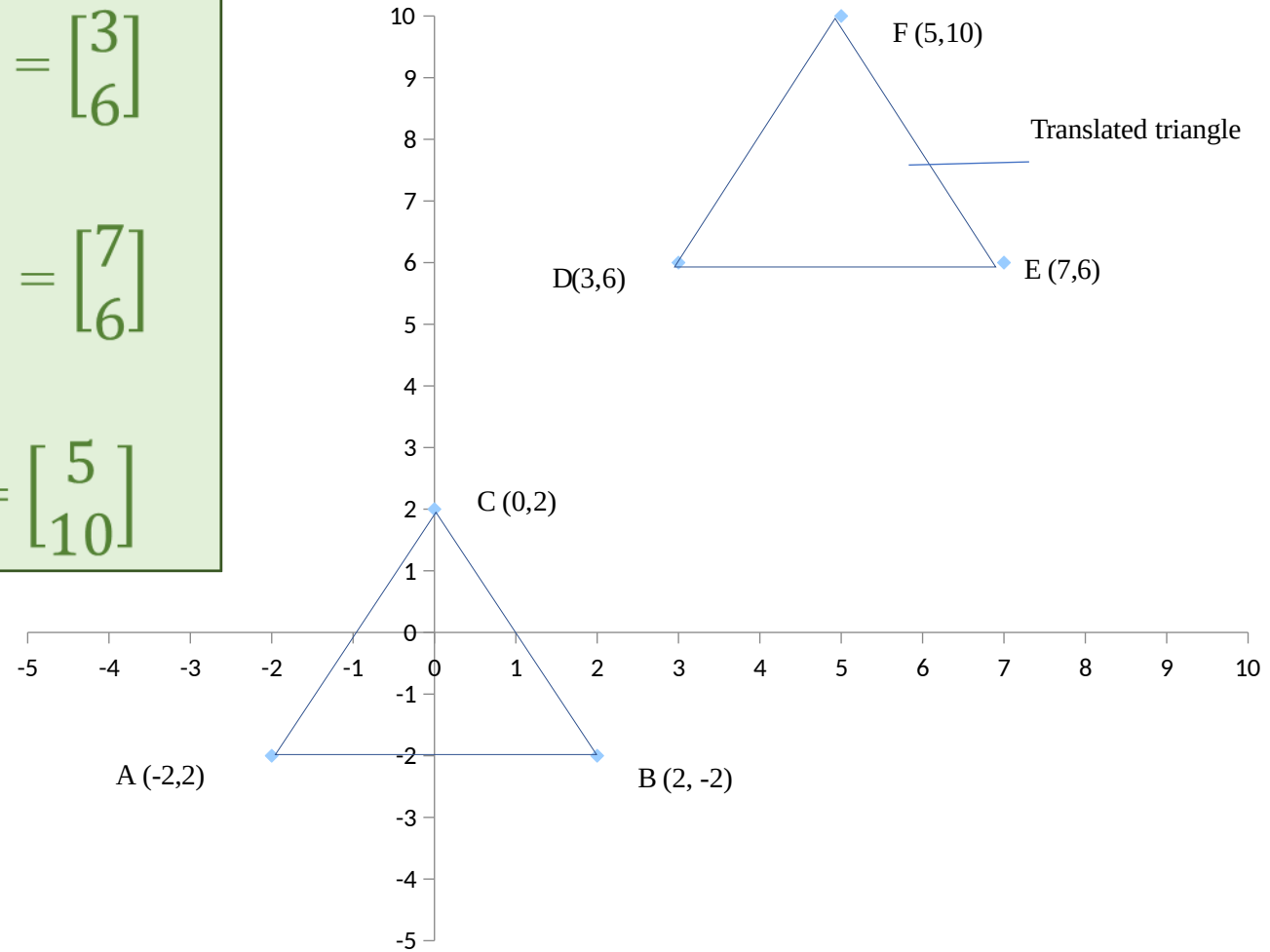
Find the translated triangle with vector $\vec{a} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$.

Example: Solution

For point A: $D = T(A) = \begin{bmatrix} -2 \\ -2 \end{bmatrix} + \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$

For point B: $E = T(B) = \begin{bmatrix} 2 \\ -2 \end{bmatrix} + \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$

For point C: $F = T(C) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$



Translation of a line

Standard form of a line: $Ax + By + C = 0$

Translation by vector $\vec{t} = (a, b)$

We want to find the **new equation** after translating all points by (a, b) .

So, each point (x, y) becomes $(x', y') = (x + a, y + b)$

Substitute backward:

From $(x', y') = (x + a, y + b)$,

we get: $x = x' - a, y = y' - b$

Substitute into the original equation: $A(x' - a) + B(y' - b) + C = 0$

Expand: $Ax' - Aa + By' - Bb + C = 0$

Group terms: $Ax' + By' + (C - Aa - Bb) = 0$

Translation of a line: Example

The **translated line** is:

$$Ax + By + (C - Aa - Bb) = 0$$

Where:

- A, B, C are the coefficients of the original line,
- a, b are the translation amounts in x – and y – directions.

Example: (Translation of a line)

For a line $3x + 4y - 5 = 0$, find the equation of line translated through vector $\vec{a} = (2, -1)$.

Solution:

Compute new constant term:

$$C' = -5 - (3)(2) - (4)(-1) = -5 - 6 + 4 = -7$$

So the new line is:

$$3x + 4y - 7 = 0$$

$$\text{Or } x' = x + 2, y' = y - 1$$

$$3(x' - 2) + 4(y' + 1) - 5 = 0$$

$$3x' + 4y' - 7 = 0$$

Translation of a Circle

General form of a circle: $(x - h)^2 + (y - k)^2 = r^2$

Translation by vector $\vec{t} = (a, b)$

We want to find the **new equation** after translating all points by (a, b) .

So, Center (h, k) becomes $(h', k') = (h + a, k + b)$ and radius does not change

The **translated circle** is:

$$(x - (h + a))^2 + (y - (k + b))^2 = r^2$$

Or more simply:

$$(x - h')^2 + (y - k')^2 = r^2$$

Where

$$h' = h + a, \quad k' = k + b$$

Translation of a Circle: Example

Example:

For a circle

$$(x - 3)^2 + (y + 2)^2 = 9$$

find the equation of circle translated through vector

$$\vec{a} = (-1, 4)$$

Solution:

Center: $(3, -2)$, Radius: 3

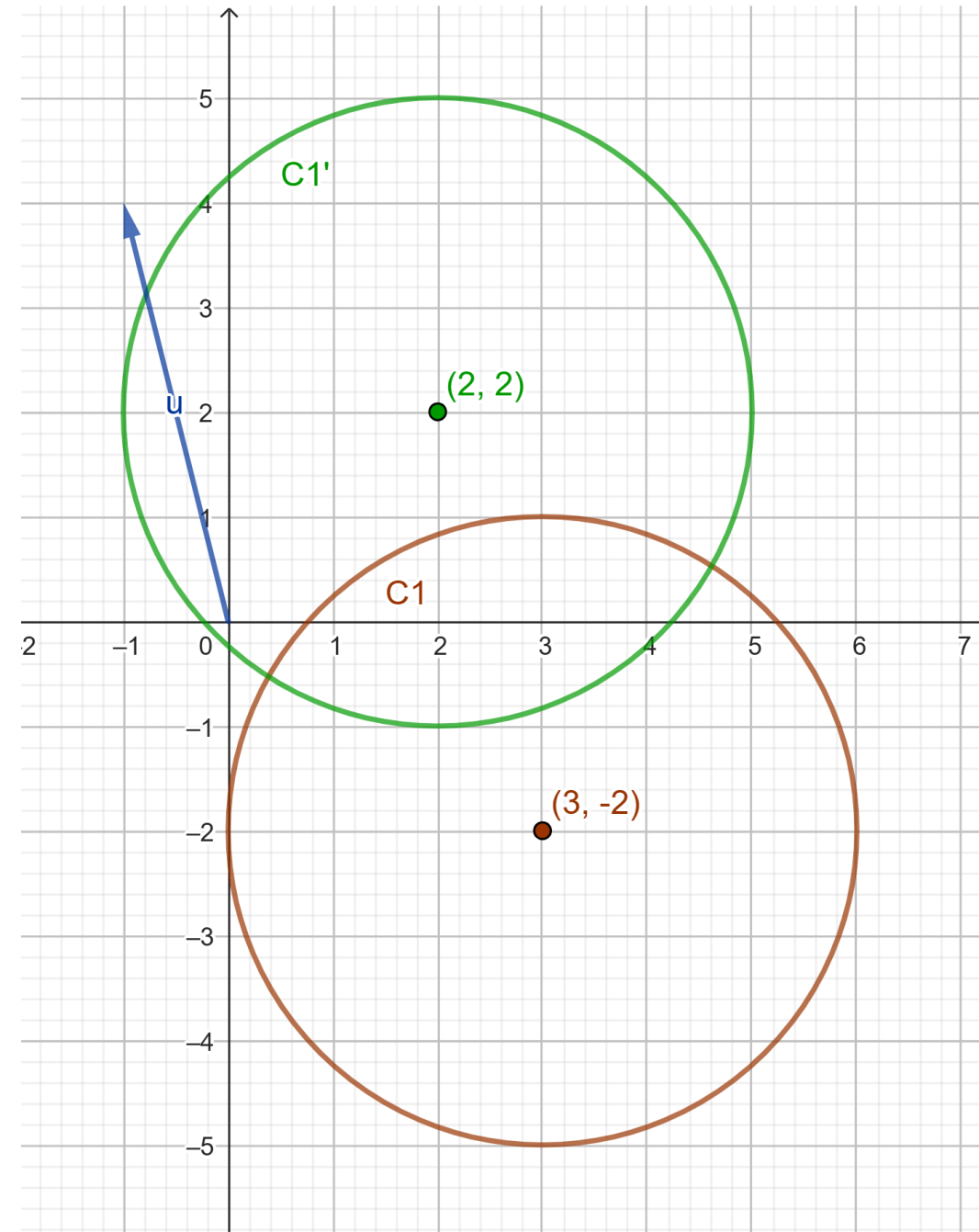
New center: $(3 - 1, -2 + 4) = (2, 2)$

New Circle is $(x - 2)^2 + (y - 2)^2 = 9$

Or $x' = x - 1, y' = y + 4$

$$(x' + 1 - 3)^2 + (y' - 4 + 2)^2 = 9$$

$$(x' - 2)^2 + (y' - 2)^2 = 9$$



Reflection (Linear Transformation)

In linear algebra, a **reflection** is a transformation that "flips" a figure over a specific line (in 2D) or plane (in 3D), creating a mirror image.

Reflection about x – axis

A **reflection about the x-axis** flips a point over the x-axis, meaning: $(x, y) \rightarrow (x, -y)$

Matrix Representation

$$R_{x-axis} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Ex. For any point $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, the Reflection is

$$\vec{v}' = R_{x-axis} * \vec{v} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

So, Original Point (3,4), Reflected Point (3, -4)

Reflection (Linear Transformation)

Reflection about y – $axis$

A **reflection about the y-axis** flips a point over the y-axis, meaning: $(x, y) \rightarrow (-x, y)$

Matrix Representation

$$R_{y-axis} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Ex. For any point $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, the Reflection is

$$\vec{v}' = R_{y-axis} * \vec{v} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

So, Original Point (3,4),
Reflected Point (−3,4)

Reflection (Linear Transformation)

Reflection about *Origin*

A **reflection about the Origin** flips a point over the Origin, meaning: $(x, y) \rightarrow (-x, -y)$

Matrix Representation

$$R_{origin} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Ex. For any point $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, the Reflection is

$$\vec{v}' = R_{origin} * v = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$$

So, Original Point (3,4), Reflected Point (-3, -4)

Reflection (Linear Transformation)

Reflection about line $y = x$

A **reflection about $y = x$** flips a point over the line, meaning: $(x, y) \rightarrow (y, x)$

Matrix Representation

$$R_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Ex. For any point $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, the Reflection is

$$\vec{v}' = R_{y=x} * \vec{v} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} * \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

So, Original Point (3,4), Reflected Point (4,3)

Reflection (Linear Transformation)

Reflection about line $y = -x$

A **reflection about $y = -x$** flips a point over the line, meaning: $(x, y) \rightarrow (-y, -x)$

Matrix Representation

$$R_{y=-x} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Ex. For any point $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, the Reflection is

$$\vec{v}' = R_{y=-x} * \vec{v} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} * \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$$

So, Original Point (3,4), Reflected Point (-4, -3)

Reflection of a line $y = mx + c$ about $y = x$

Since reflecting over $y = x$ means switching coordinates, we swap x and y in the original equation:

$$y = mx + c \rightarrow x = my + c \Rightarrow y = \frac{1}{m}x - \frac{c}{m}$$

Example. Reflect $y = 2x + 5$ over the line $y = x$

$$x = 2y + 5$$

$$2y = x - 5$$

$$y = \frac{1}{2}x - 5/2$$

Reflection of Points and Line about line $y = x$

$$(x, y) \rightarrow (y, x) \text{ And}$$

$$R_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Points

$$A(-3, 4), B(-4, 2), (-1, 3)$$

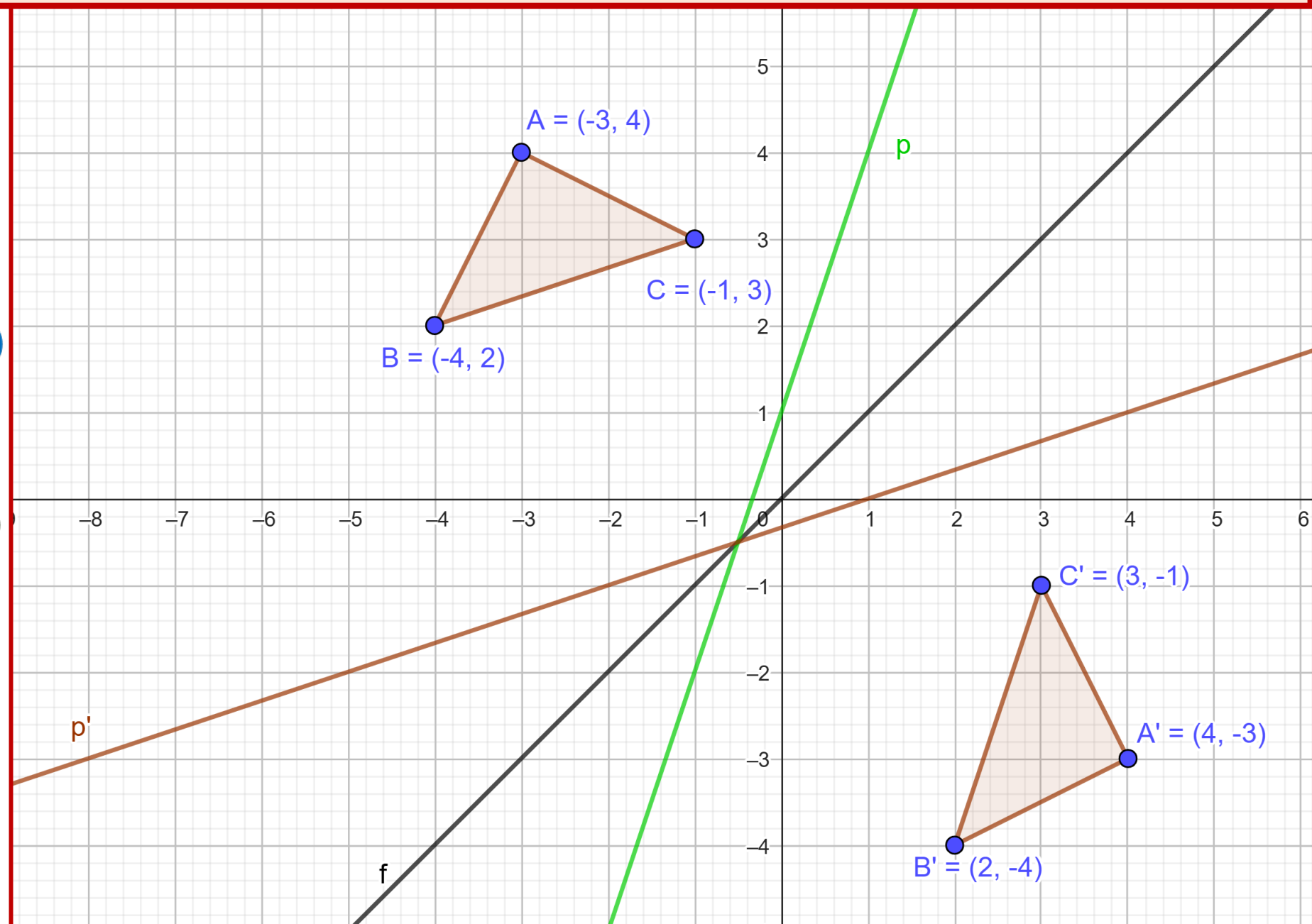
Transformed Points

$$A(-3, 4), B(-4, 2), (-1, 3)$$

$$\text{Line: } y = 3x + 1$$

$$\text{About: } y = x$$

$$\text{Transformed Line:} \\ x - 3y = 1$$



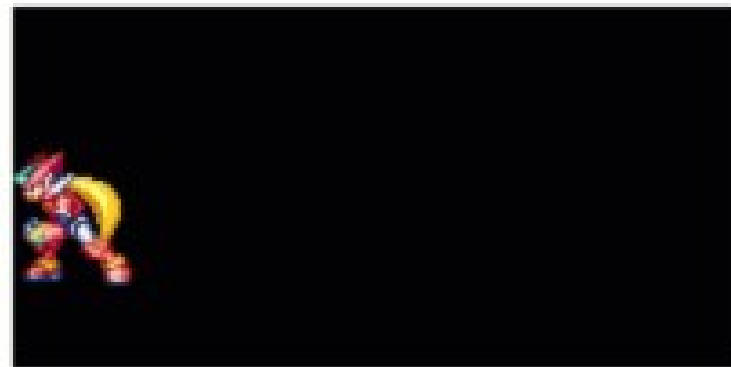
Home Work:

Verify other cases on GeoGebra

Real-world scenario: Reflection in Game Development or UI design

You're designing a **2D platformer game** or a **graphical user interface (GUI)**.

Suppose you have a character or object sprite walking to the right, and now you want it to **turn around and walk to the left** — **without redrawing a new sprite**.



Rotation (Linear Transformation)

Any point (x, y) can be represented as:

$$x = r * \cos(\phi), \quad y = r * \sin(\phi)$$

- r is the distance from the origin:
- ϕ is the angle point makes with the x – axis

After rotating by θ ,
the **new angle becomes $\phi + \theta$** .

So, the new coordinates (x', y') are:

$$x' = r * \cos(\phi + \theta)$$

$$y' = r * \sin(\phi + \theta)$$

$$x' = r \cos \phi \cos \theta - r \sin \phi \sin \theta$$

$$y' = r \sin \phi \cos \theta + r \cos \phi \sin \theta$$

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

Matrix Form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation (Linear Transformation)

So, Rotation Matrix is

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

This rotates any 2D vector by θ anticlockwise about the origin.

And Rotation Matrix in Clockwise direction

$$R(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

Summary Table

Rotation Matrix	
Angle θ	Rotation Matrix
$60^\circ \left(\frac{\pi}{3}\right)$	$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$
$90^\circ \left(\frac{\pi}{2}\right)$	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
$180^\circ (\pi)$	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
$270^\circ \left(3\frac{\pi}{2}\right)$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
Angle θ	Rotation Matrix
$60^\circ \left(\frac{\pi}{3}\right)$	$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$
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$180^\circ (\pi)$	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
$270^\circ \left(3\frac{\pi}{2}\right)$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Example 1: Sketch the image of given rectangle with vertices A(0,0), B(3,0), C(3,2), D(0,2) under the rotation of 30° (anticlockwise)

Solution: As the transformation of rotation is

$$T(\vec{x}) = A\vec{x} \quad : \forall \vec{x} \in R^2$$

Where

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

As $\theta = 30^\circ$, so

$$A = \begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\text{Point A: } T \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{Point B: } T \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 0 \end{bmatrix}$$

$$\text{Point C: } T \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 3.2 \end{bmatrix}$$

Example 1: Sketch the image of given rectangle with vertices A(0,0), B(3,0), C(3,2), D(0,2) under the rotation of 30°

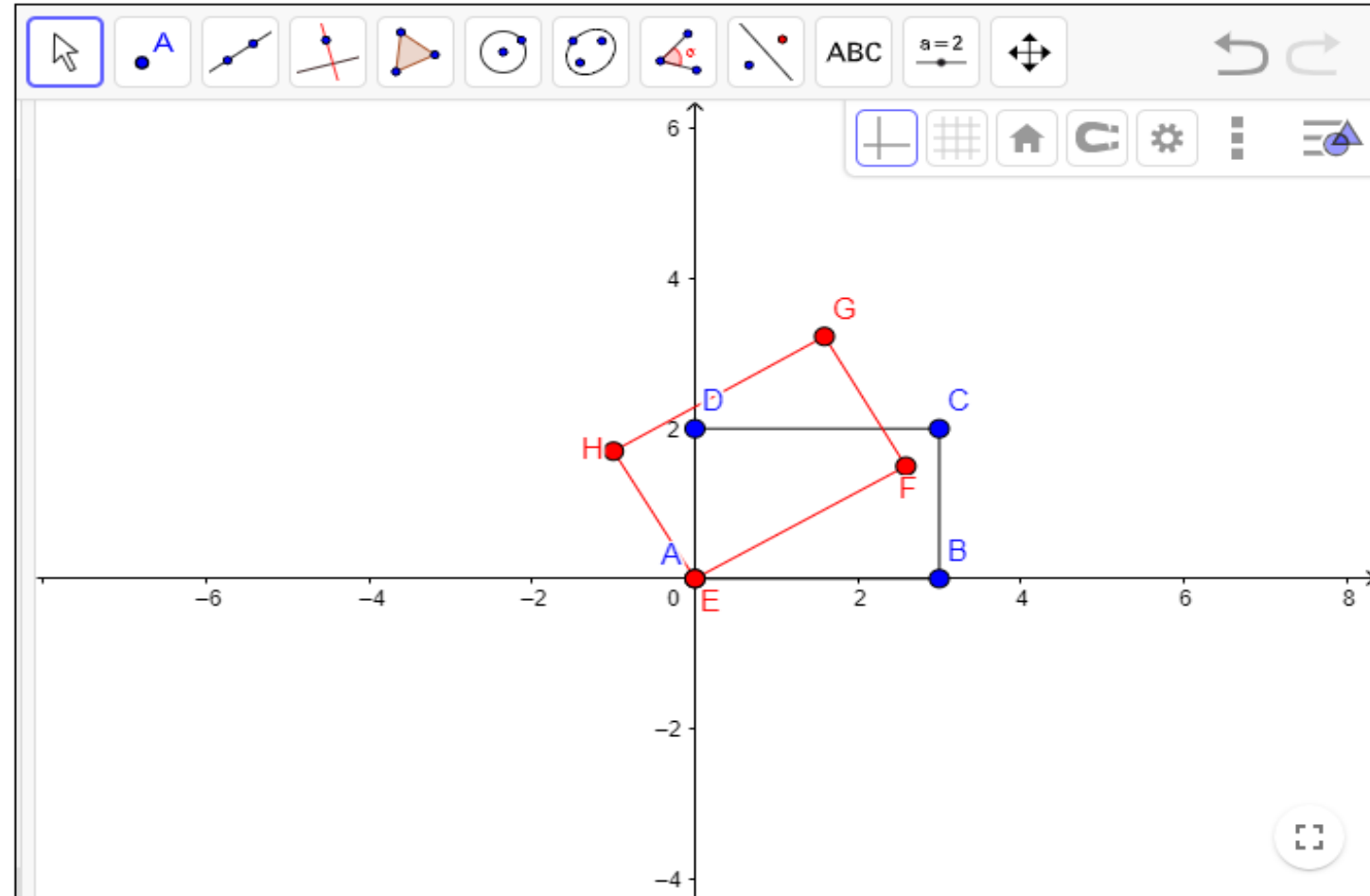
$$\text{Point } D: T \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 1 & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1.7 \end{bmatrix}$$

So, the vertices

A(0,0), B(3,0), C(3,2), D(0,2)

are now transformed

E(0, 0), F(2.6, 0), G(1.6, 3.2), H(-1, 1.7)



Example: Let $(x - 4)^2 + (y - 3)^2 = 9$ be a circle. Find the equation of circle after rotating it through an angle of 90° in anticlockwise direction about origin.

Solution: As the transformation of rotation is

$$T(\vec{x}) = A\vec{x} \quad : \forall \vec{x} \in R^2$$

And Matrix with angle $\theta = 90^\circ$

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

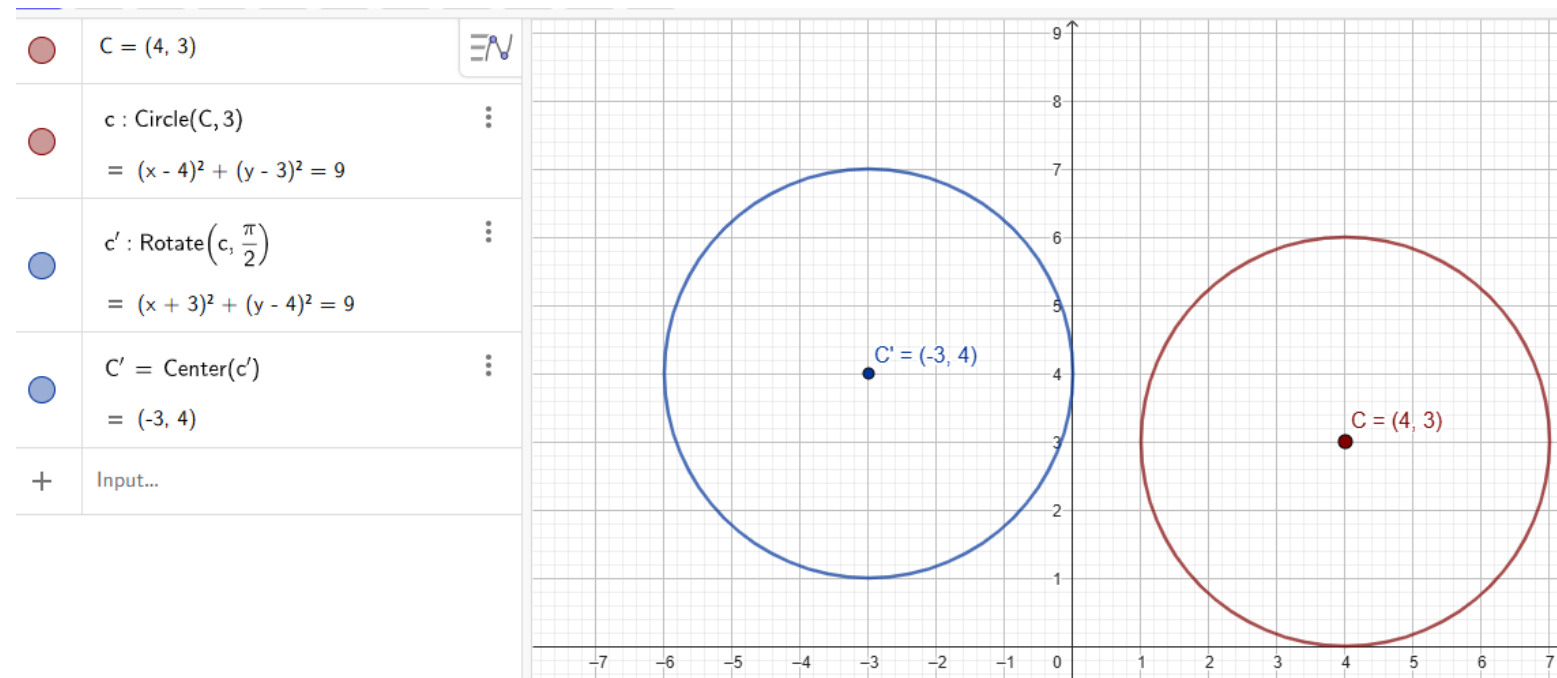
$$\Rightarrow x = y' \text{ And } y = -x'$$

Putting values of x and y in original equation of circle, we get

$$(y' - 4)^2 + (-x' - 3)^2 = 9$$

Or

$$(x' + 3)^2 + (y' - 4)^2 = 9$$



Example 1: Sketch the image of given rectangle with vertices A(0,0), B(3,0), C(3,2), D(0,2) under the rotation of 30°

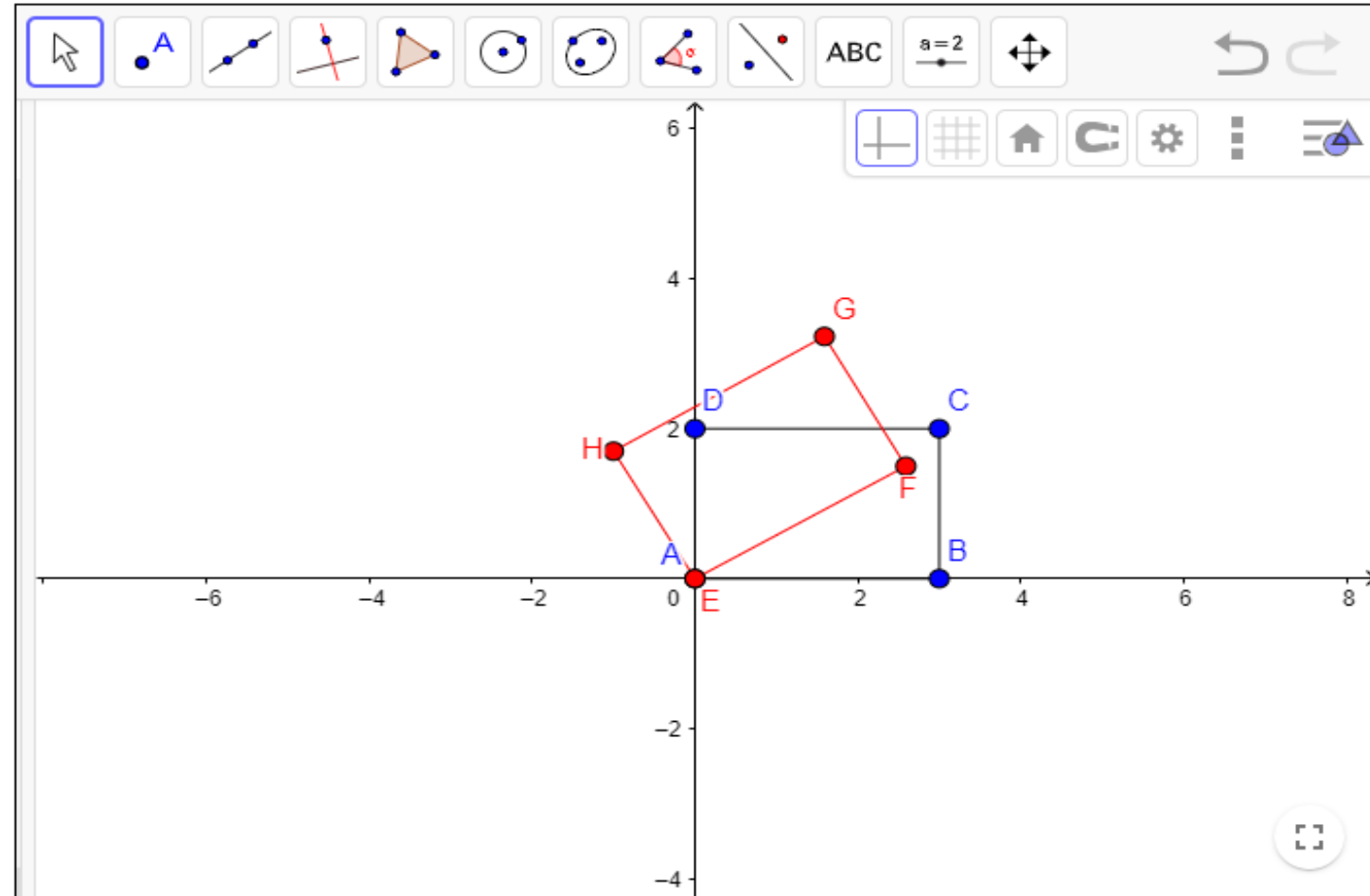
$$\text{Point D: } T \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1.7 \end{bmatrix}$$

So, the vertices

A(0,0), B(3,0), C(3,2), D(0,2)

are now transformed

E(0, 0), F(2.6, 0), G(1.6, 3.2), H(-1, 1.7)



Home Work:

Exercise in Word File

9. Matrix Transformation

10. Euclidean Transformation

THANKS